Wheat Time Series Analysis

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1 Introduction

One of the principal staple crops, wheat is the primary source of nutrition for billions of people worldwide. It is very significant in various fields, such as international trade, economics, agriculture and food production, and has emerged as a symbol and epitome of the state of nourishment worldwide. Understanding changes in wheat prices and how they fluctuate during months and more broadly years is fundamental: both producers and consumers nationwide must be aware of the variations and shifts since the prices affect agricultural policies and the whole global economy. Wheat prices exhibit deep dynamics that are impacted by multiple variables, like geopolitical events, demand and supply relations, technological developments, trade laws and market speculation. In our project, we recognise those prices as a stochastic process with inherent uncertainty and variability caused by the factors listed above and we use autocorrelation function (ACF) and partial autocorrelation function (PACF) plots, together with the fitting of an autoregressive integrated moving average (ARIMA) process, in order to hit our main target: gaining insight into the complex trends and swings in the price of wheat over time. Moreover, we aim to undercover patterns and key drivers and investigate potential factors that influenced the price of wheat by analyzing around 34 years of data given by the International Monetary Fund and available on FRED (Global price of Wheat), using as the variable of interest a Metric Ton in U.S. Dollars and a monthly frequency to collect the observations. Data start from January 1990 and the last data is gathered in May 2023.

2 Historical Analysis

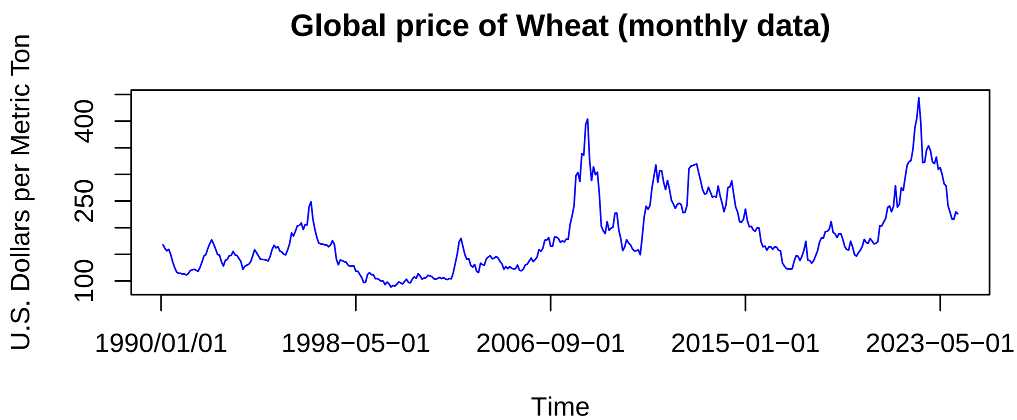

Figure 1: Global price of Wheat (monthly data)

```
#general representation of time series graphically
t = read.table("Wheat.txt", sep=",")
y = ts(t[, 2])
plot(y, col="blue", xaxt = "n", ylab="U.S. Dollars per Metric Ton", xlab="Time",
    main="Global price of Wheat (monthly data)" )
observational_x_val <- c(0, 100, 200, 300, 400)
custom_values <- c("1990/01/01", "1998-05-01", "2006-09-01", "2015-01-01",
    "2023-05-01")
axis(1, at = observational_x_val, labels = custom_values)
```

Given the detailed data we have available, we are able to comment how and when the global price of wheat changed and for what reasons. Starting from the beginning, from 1990 to 1995 the series did not encounter peaks or lows worth considering and there was not much volatility overall. The stability in prices can be attributed to a lack of important political or environmental events. On the other hand, in 1996 the price of wheat rose up to almost 250 dollars per metric ton. The spike during this period can be attributed to a mix of factors: firstly, two main conflicts such as the Bosnian War and the Chechen War disrupted key wheat-producing countries, leading to supply decreasing. Additionally, severe droughts in countries like Australia (in the top 10 for most produced wheat) and in the United States (ranked fifth) impacted the production. Following a period of wheat surplus and stocks, there was a protracted period of generally low prices. In addition, in 1997 the south-east region of Asia (especially Thailand, Philippines and Indonesia) were struck by a critical financial crisis that led to a decrease in the demand of wheat by countries that historically utilize the crop and a consequent decrease in its price. The conditions of economic stability and balanced supply-demand dynamics continued to last until 2008, only disturbed in 2002 where the price of wheat increased probably as a result of the terrorist attack that took place in Manhattan on the 11 of September 2001. The attack, masterminded by Al-Qaeda, resulted in the killing of 2977 people and in a subsequent heavy cultural and financial crisis that led to the general increase of necessities' prices and a broader economic uncertainty. In 2008, the world was struck by the most important financial crisis after the Great Depression of 1929. The graph shows clearly the seriousness of the catastrophe displaying the price of wheat that more than doubled in only 10 months, reaching an all-time high until then of 403 U.S. dollars. The crisis, which affected all Western-influenced countries, impacted on the price of staple goods for several reasons: the shift of consumers' behaviour, which cut discretionary spending and concentrated on buying necessities and this raised prices in many fundamental sectors. Moreover, in response to the financial crisis, central banks adopted expansionary monetary policies such as quantitative easing and interest rate reductions. These measures have certainly increased inflation and as a consequence the cost of needs. The wheat market never fully recovered from 2008, with prices that show high volatility that continued up to 2015 and a mean cost that is far from what it used to be before the crisis. That could be attributed to both the effects of the crisis that still went on, aggravated by the recessions of Greece, Spain and Italy (big wheat producer), and to the instability of Middle East countries where several internal conflicts blew up and mined wheat production and its market. After a brief decrease that went on from 2016 to 2020 and still showed medium variability, the wheat market entered its most critical period. In 2020, the world was struck by one of the most serious natural emergencies in the Modern Era. The effects of the pandemic did not only weaken both mental and physical health, but also the health of the global economy. After a year that saw economic slowdown and a worldwide stagnation, COVID-19 lockdowns and other precautions in order to limit the spread of the virus drove the global economy into another crisis. Wheat, as much as other fundamental goods, suffered a steady increase of its cost due to various factors: along with the general depreciation of almost every global currency, the regulations applied to stop the spread of the pandemic caused the amount of wheat available on the market to fall down. Both because of the limited possibility to harvest it and transport it, and because of the general panic buying and stockpiling that saw pasta, bread and other wheat products as the most purchased. Subsequently in 2022 the wheat market in particular suffered another peak, at 444 U.S. dollars which is the all-time high based on the data available from 1990, as a result of the invasion of Ukraine by Russia. Ukraine, ninth leading wheat producer and trader, saw its crop market heavily affected by the Russian military and government as a means of weakening its economy that highly relies on the sale of wheat to especially Southern-Asian and Middle Eastern countries. The invader military targeted dams with heavy bombardments in the South regions of Ukraine, flooding fields and damaging agriculture particularly in Kherson. Moreover, attacks and interdiction on key Ukrainian ports along the Black Sea, such as Odessa and Mariupol, caused significant damage to the country's export infrastructure, and thus, its capacity to ship wheat to foreign markets. By July 2022 Turkish-mediated negotiations between Ukraine and Russia led to an agreement under which the Russian blockade of Ukraine's Black Sea ports would be lifted and the safe passage of grain shipments guaranteed. This initiative has been key in mitigating the high prices of wheat, which steadily decreased and have continued to do so until February 2024.

3 Preliminary forecasting

In the following section, employing methods of time series analysis, patterns can be unveiled and underlined. The primary goal is to find stationarity and seasonal patterns to reproduce and try to predict the series.
Here are several key points to consider when analyzing the series:

3.1 Logarithmic transformation

Given the upward trend observable in Figure 1, a smart approach is to mitigate the increasing trend by taking the logarithmic transformation of the observations. The method allows an attenuation of fluctuations, especially considering the higher peaks in 2008 and 2022, with the ultimate goal of getting closer to stationarity.

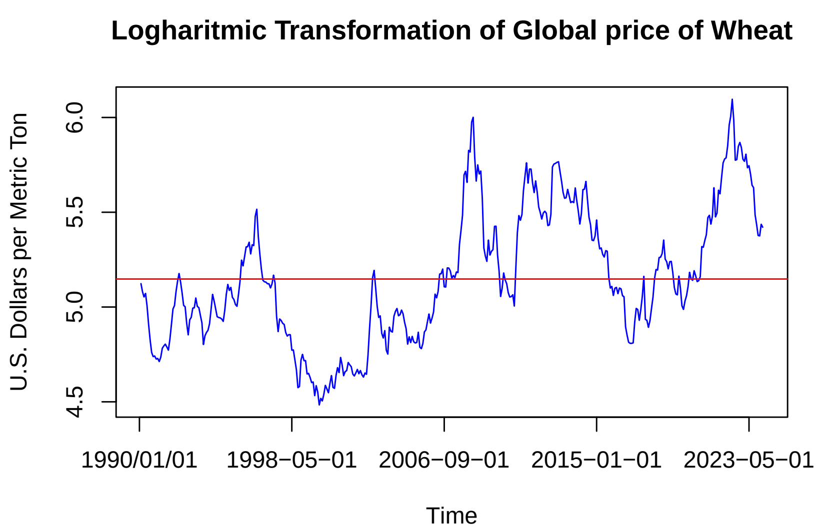

Figure 2: Logarithmic Transformation of Global price of Wheat

```r
#log scaled transformation
ly = log(y)
plot(ly, col="blue", xaxt = "n", ylab="U.S. Dollars per Metric Ton", xlab="Time",
    main="Logharitmic Transformation of Global price of Wheat" )
axis(1, at = observational_x_val, labels = custom_values)
abline(h=mean(ly), col="red")
```

As we can see, the data got a lot closer to the mean of the time series, that is drawn in the graphical representation by the red line, at the value 5.15 dollars.

3.2 ACF and PACF

Understanding the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of a time series is fundamental for forecasting. These functions not only allow the analysis of patterns, seasonality, and influencing factors but also play a crucial role in the preliminary analysis of a time series by providing insights into the potential influences of Autoregressive (AR) and Moving Average (MA) components. By examining ACF and PACF plots, analysts can gain an understanding of how these components contribute to the observable patterns in the data and inform the selection of appropriate forecasting models such as ARIMA (Autoregressive Integrated Moving Average).

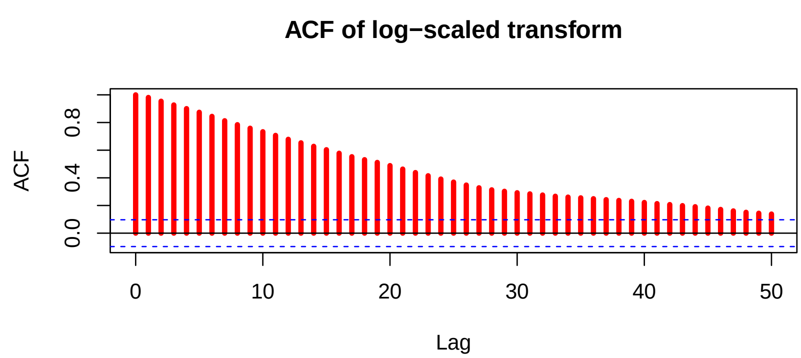

Figure 3: ACF of log-scaled transform

```r
#ACF plotted
acf(ly, lag.max=50, col="red", lwd=4, main="ACF of log-scaled transform")
```

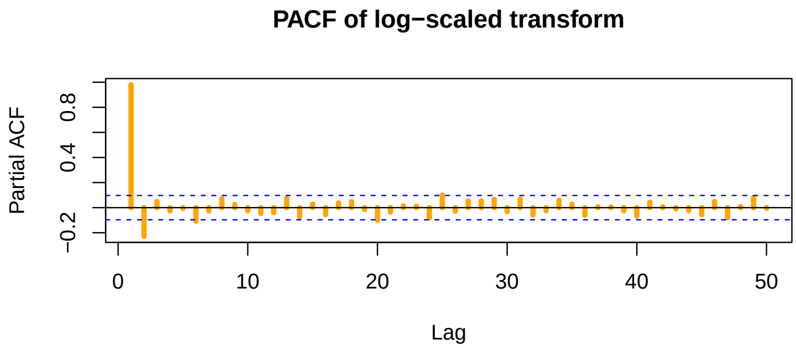

Figure 4: PACF of log-scaled transform

```
#PACF plotted
pacf(ly, lag.max=50, col="orange", lwd=4, main="PACF of log-scaled transform")
```

Doing a preliminary analysis, we can immediately observe from the plot of the Autocorrelation function that it likely exhibits a significant Autoregressive component. This is indicated by the slow, linear decrease in the Autocorrelation values, suggesting a strong influence of past values on future ones. As expected, from the analysis of the Partial Autocorrelation Function, we can discern that the model is not purely Autoregressive. If it were, we would expect the Partial Autocorrelation to be null after a certain lag, depending on the order of the Autoregressive component. However, this is merely the initial interpretation based on the ACF and PACF plots. For further analysis, achieving stationarity through differencing techniques is mandatory.

3.3 Differenced log-scaled transform

When performing time series analysis, the differenced log-scaled transformation is an effective tool that may be used to display the data and for representing the model in an easier way. The new series illustrates the percentage changes between consecutive observations, which shed light on the underlying dynamics of growth or decay. It is taken to stabilize variance and to make exponential trends additive. Moreover, the resulting log-scaled transform graph often exhibits stationary properties, making it suitable for further analysis.

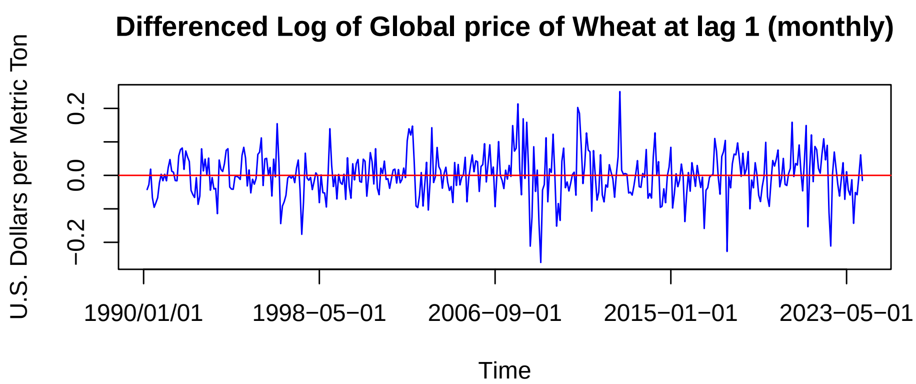

Figure 5: Differenced log-scaled transform of Wheat at lag 1 (monthly)

```
#differenced log-scaled
dly = diff(ly)
plot(dly, col="blue", xaxt = "n", ylab="U.S. Dollars per Metric Ton", xlab="Time",
    main="Differenced Log of Global price of Wheat at lag 1 (monthly)" )
axis(1, at = observational_x_val, labels = custom_values)
abline(h=0, col="red")
```

The relatively steady statistical properties of the investigated time series across time indicate characteristics suggestive of stationarity. This seems to be true despite a few peaks (i.e. 2008, 2022), suggesting that both mean and variance have not changed systematically and even though they are evident, these isolated peaks do not contrast substantially from the overall pattern that we can observe from our data. To ensure stationarity and determine whether another differencing step is necessary, which could result in data loss, we can conduct the Augmented Dickey-Fuller (ADF) test in R to obtain the p-value.

```
   #ADF test
    library(tseries)
adf.test(dly)
```

Performing the test, we get a statistic value of -7.598 with a lag order of 7 corresponding in a p-value smaller than 0.01, which is enough to be highly confident that we achieved stationarity.

3.4 Seasonality

Another topic of preliminary forecasting involves investigating seasonal patterns at various lags. One hypothesis we consider is that the fluctuations in the price of wheat can be examined across different years (lag 12) or seasons (lag 4). This hypothesis is based on the conjecture that crops may be influenced by significant events occurring in specific years or by varying conditions across different seasons.

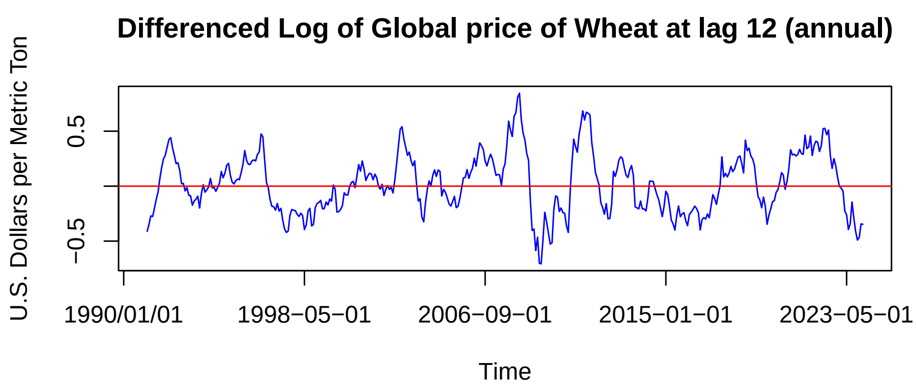

Figure 6: Differenced Log of Global price of Wheat at lag 12 (annual)

```
#12 seasonal difference
dsly_12 = diff(ly, 12)
plot(dsly_12, col="blue", xaxt = "n", ylab="U.S. Dollars per Metric Ton",
    xlab="Time", main="Differenced Log of Global price of Wheat at lag 12
    (annual)" )
axis(1, at = observational_x_val, labels = custom_values)
abline(h=0, col="red")
```

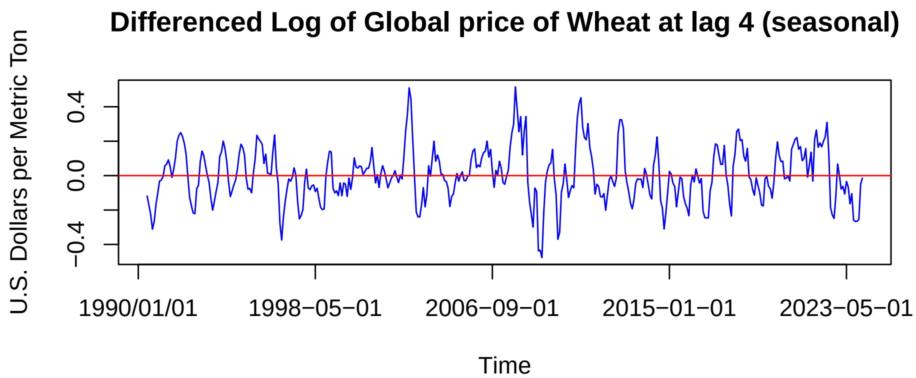

Figure 7: Differenced Log of Global price of Wheat at lag 4 (seasonal)

```
#4 seasonal difference
dsly_4 = diff(ly, 4)
plot(dsly_4, col="blue", xaxt = "n", ylab="U.S. Dollars per Metric Ton",
    xlab="Time", main="Differenced Log of Global price of Wheat at lag 4
    (seasonal)" )
axis(1, at = observational_x_val, labels = custom_values)
abline(h=0, col="red")
```

As we can observe visually, both of the graphs oscillate around the red line with a similar overall frequency, suggesting that the differencing techniques likely achieved stationarity. However, the seasonal components are unlikely to be predominant because if they were, we would have observed some form of seasonal recurrent pattern in the ACF (Figure 3). Moreover, it is preferable to consider the data with the monthly difference as it provides the most favorable dataset, as confirmed by applying various functions in R. These functions include Standard Deviation, which measures overall variability; Range, used to identify outliers; Skewness, a measure of asymmetry around the mean; and Kurtosis, which provides insight into the peakedness of the distribution.

```
    #TEST FOR VARIABILITY FOR DIFFERENT DIFFERENCED LAGS FUNCTION:
meas_variability <- function(ts) {
  sd <- sd(ts)
  range <- max(ts) - min(ts)
  skewness <- moments::skewness(ts)
  kurtosis <- moments::kurtosis(ts)
  results <- data.frame(Method = c("Standard Deviation", "Range", "Skewness",
      "Kurtosis"),
  Value = c(sd, range, skewness, kurtosis))
```

```
  return(results)
}
#test for dly
variability_dly = meas_variability(dly)
#test for dsly_12
variability_dsly_12 = meas_variability(dsly_12)
#test for dsly_4
variability_dsly_4 = meas_variability(dsly_4)
```

Differenced Wheat TS	Standard Deviation	Range	Skewness	Kurtosis
Monthly difference (lag 1)	0.067	0.510	-0.00132	4.488
Annual difference (lag 12)	0.269	1.553	0.258	2.798
Seasonal difference (lag 4)	0.155	0.992	0.21	3.52

Table 1: Variability Measures for Different Differenced data

As we can see by the values in the table, it is possible to highlight that the monthly differenced data of the wheat is preferable, because it has a lower range and standard deviation and it has a skewness very close to 0, meaning that it has a good symmetry. Although, the seasonal and annual differenced data have both a lower kurtosis, meaning that they share fewer outliers respect to the monthly differenced data set.

4 Estimation of the model

The purpose of the following section is to find suitable models that can accurately reflect the distribution and its properties. We will conduct a series of simulations of ARIMA models that could potentially be appropriate for the process. In previous sections, we determined that taking a first-order difference of the logarithmic distribution of wheat is adequate to achieve stationarity and establish a supervised process. In fact, we exclude the presence of a strong seasonal component by the considerations done before. Through the analysis of the Autocorrelation function (ACF) of the logarithmic first-order differenced distribution of wheat, we can gain insights into processes that are not suitable.

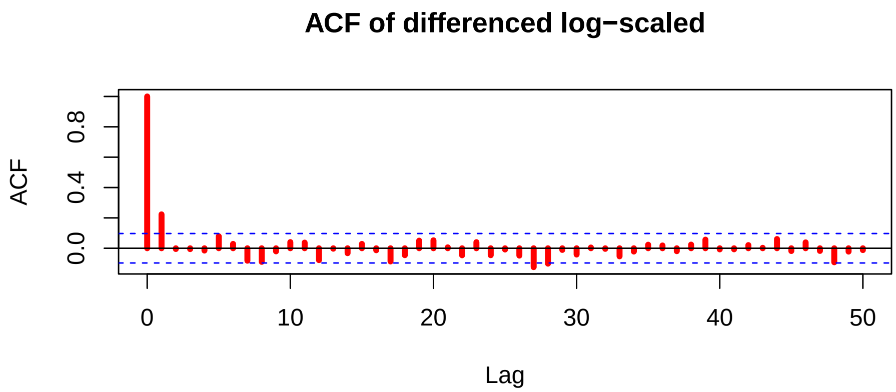

Figure 8: ACF of differenced log-scaled

From figure 8, it is clear that the ACF never gets to 0 and exhibits a sinusoidal behaviour. From this information, we can rule out "a priori" an exclusively Moving Average component and infer some Autoregressive conduct. Moreover, through the analysis of the ACF, it suggests incorporating a MA(1) component in the ARIMA estimation, as evidenced by the significant decrease of Autocorrelation after the first lag.

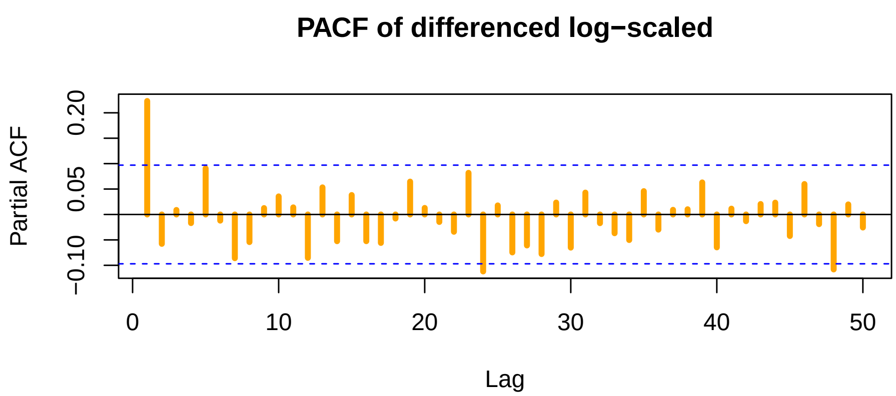

Figure 9: PACF of differenced log-scaled

Examining figure 9 reinforces our beliefs that a Moving Average component is evident, otherwise the PACF would reach 0 after a certain lag. At this point, we understand that simulating the log-differenced time series requires an ARIMA(p,d,q) process. However, as demonstrated in Section 3.3, stationarity has already been achieved, so additional differencings are not needed. Hence empirical estimation can be done, under the conjecture of

an ARIMA(p,0,q) where both parameters p and q are non-zero. Starting from the simplest process to simulate, consider an ARIMA(1,0,1) model with parameters ϕ for the AR(1) and θ for the MA(1).
We'll adopt σ=0.0671, which is the most reasonable standard deviation to use for the White Noises ϵ_t, given that the standard deviation of the differenced log-scaled data is equal to this value.

```
#Standard Deviation of the series
standard_deviation <- sd(dly)
```

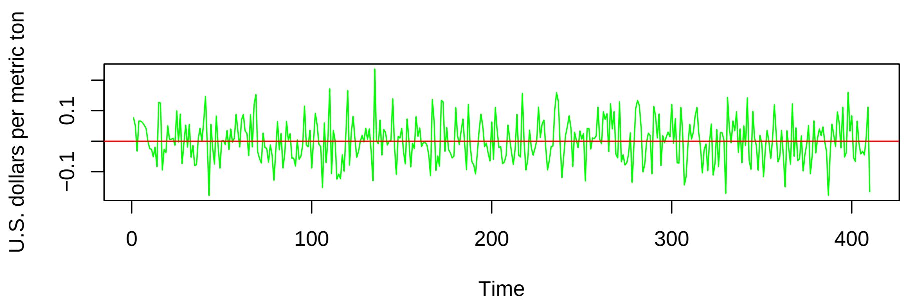

Figure 10: Simulation of an ARIMA(1,0,1) with ϕ=-0.8 and θ=1

Anyway, despite models like the one depicted in Figure 10 resembles very much the behaviour of the studied time series, the best results were achieved considering ARIMA(2,0,1) processes across various different seeds with θ=1.

The most accurate simulation across multiple seeds has been found by setting the ARIMA(2,0,1) with the following parameters: ϕ_1=-0.7, ϕ_2=0.2 and θ=1; without any shift. The ACF and the PACF exhibits very similar behaviours with respect to the ACF and PACF of the analyzed wheat's time series, that is a crucial prerequisite to ensure that the simulation is an accurate representation of the time series. It is worth noticing that while simulations are indispensable for understanding a stochastic process, data, such as global price of wheat, are often heavily influenced by worldwide events, as evidenced in figure 5. Even though data sets have been normalized, peaks in 2008 endure because of not just unpredictable White Noises, but also because of historical occurrences. A smart idea, as discussed later on, would be attempting to mitigate fluctuations linked to significant historical moments.

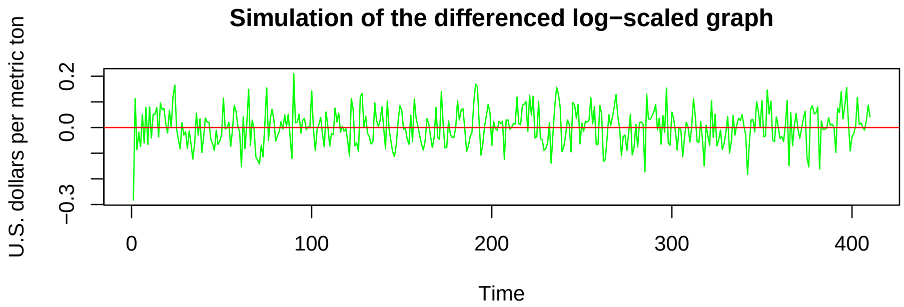

Figure 11: Simulation of ARIMA(2,0,1) with ϕ_1=-0.7, ϕ_2=0.2, θ=1, ω=0

```
#SIMULATION OF ARIMA(2,0,1)
set.seed(123)
ar2.model = list(ar = c(-0.7, 0.2))
ma1.model = list(ma = 1)
arima_sim1 = arima.sim(n=410, model=c(ar2.model, ma1.model), innov=rnorm(n=410,
    mean=0, sd=0.0671))
ts.plot(arima_sim1, main="Simulation of the differenced log-scaled graph",
    ylab="U.S. dollars per metric ton", col="green")
abline(h=0, col="red")
```

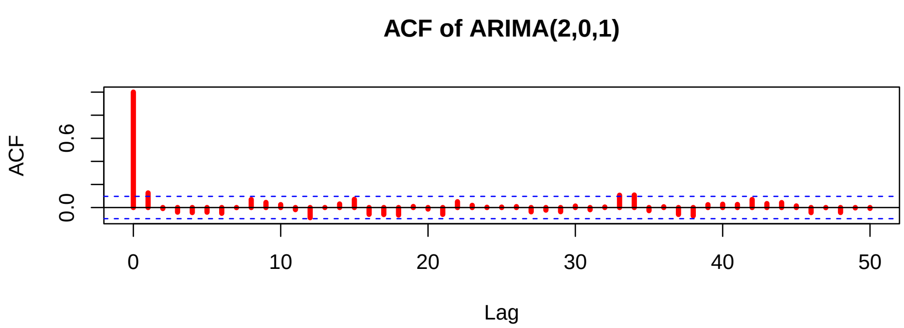

Figure 12: ACF of ARIMA(2,0,1) with ϕ_1=-0.7, ϕ_2=0.2, θ=1, ω=0

5 Spectral Density Function

Going forward with the analysis, another fundamental tool is the Spectral Density Function. With this function, we go beyond the time frame and we enter the frequency domain. Defined as the Fourier transform of the Autocovariance function, the SDF tells us how much of the overall variability of a given stochastic process is due to components associated with different frequencies, and hence different periods. Additionally, the SDF helps us in assessing stationarity and identifying anomalies, improving the accuracy of the analysis.

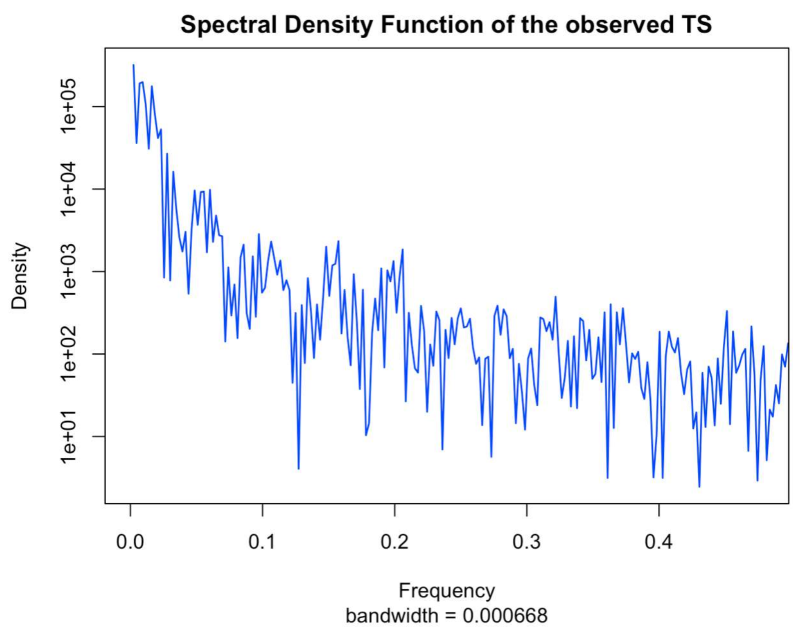

Figure 13: Spectral Density Function of the observed time series

The function is higher at the beginning and tends to decrease towards the end of the time series. This indicates that there is more variability associated with the low frequencies, that those frequencies contribute more to the overall variability than medium and high ones and that there are long periods and long term components.

```
#Spectral Density Function of the observed time series
spec=spec.pgram(y, taper=0, detrend=F, plot=F)
plot(spec, main="Spectral Density Function of the observed TS", xlab="Frequency",
    ylab="Density", type="l", col="blue", lwd=1.3, xlim= c(0, max(0.479)))
```

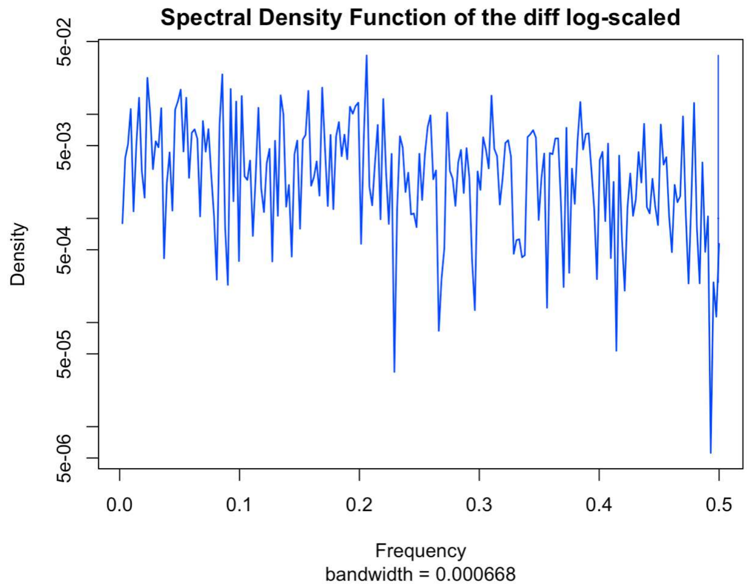

Figure 14: Spectral Density Function of the differenced log-scaled time series

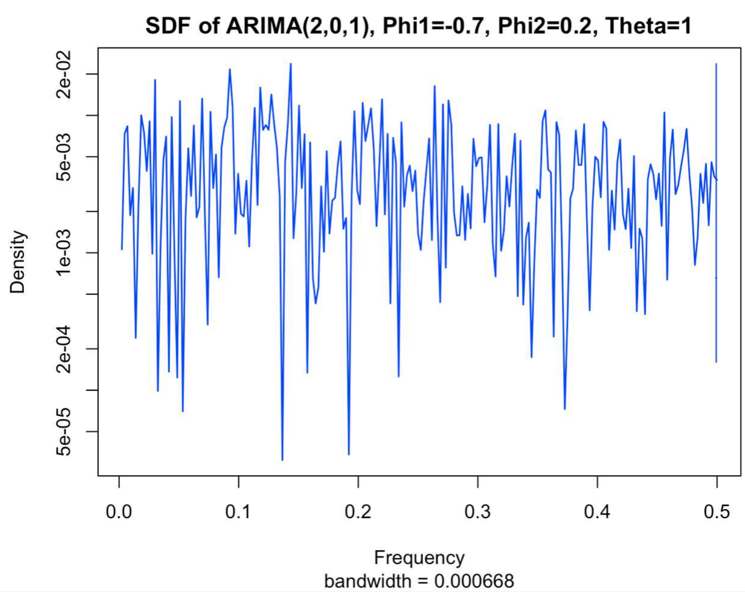

Figure 15: Spectral Density Function of the simulated ARIMA(2,0,1) with given parameters

```
#Spectral Density Function of the differenced log-scaled transform
spec=spec.pgram(dly, taper=0, detrend=F, plot=F)
plot(spec, main="Spectral Density Function of the diff log-scaled",
    xlab="Frequency", ylab="Density", type="l", col="blue", lwd=1.3)
#Spectral Density Function of the ARIMA(2,0,1) model
set.seed(123)
ar2.model = list(ar = c(-0.7, 0.2))
ma1.model = list(ma = 1)
arima_sim1 = arima.sim(n=410, model=c(ar2.model, ma1.model), innov=rnorm(n=410,
    mean=0, sd=0.0671)
spec=spec.pgram(arima_sim1, taper=0, detrend=F, plot=F)
plot(spec, main="SDF of ARIMA(2,0,1), Phi1=-0.7, Phi2=0.2, Theta=1",
    xlab="Frequency", ylab="Density", type="l", col="blue", lwd=1.3)
```

The two Spectral Density Functions above show several remarkable similarities. In the low frequencies, the density of the differenced log-scaled time series starts almost the same as the simulated one, while approaching the medium frequencies the ARIMA(2,0,1) shows more variability in its density with lower peaks than the observed one. Entering the medium frequencies, the graphics become much more comparable, almost identical in some parts (i.e. 0.3-0.4) and with resembling peaks. That is true for also the high frequencies, in consideration of the fact that the functions increase and decrease with a nearly equivalent pattern. The Spectral Density Function of the differenced log-scaled time series also reasonably confirms the idea of stationarity that was suggested beforehand while analyzing the Autocovariance function, since the SDF remains (excluding mainly one low peak) relatively stable over the whole frequency spectrum.

6 Preliminary analysis of the time series without peaks

The time series that we have analyzed, made by the Global price of Wheat from 1990 to 2024, displays two major peaks in its domain. Following the historical analysis of the process, in 2008 and in 2022 the price of wheat has steadily increased due to the Financial Crisis of 2007-2008 and due to the invasion of Ukraine by Russia. In this section, we attempt to briefly analyze the whole process excluding 2008 and 2022 peaks.

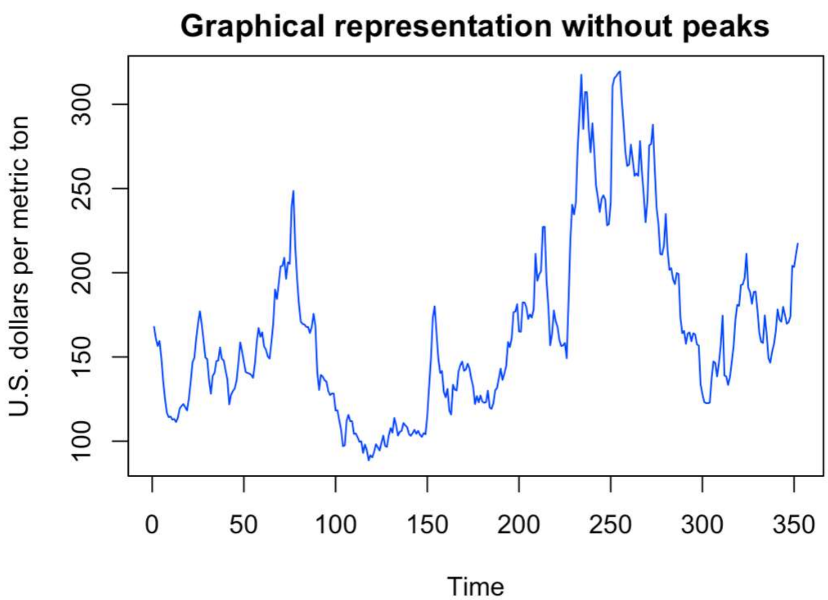

Figure 16: Graphical representation of the time series excluding 2008 and 2022

```
#Simulation of the process without 2008 and 2022 peaks
j = read.table("Wheat.no.peaks.txt", sep=",")
v = ts(j[,2])
ts.plot(v, main="Graphical representation without peaks", ylab="U.S. dollars per
    metric ton", col="blue")
```

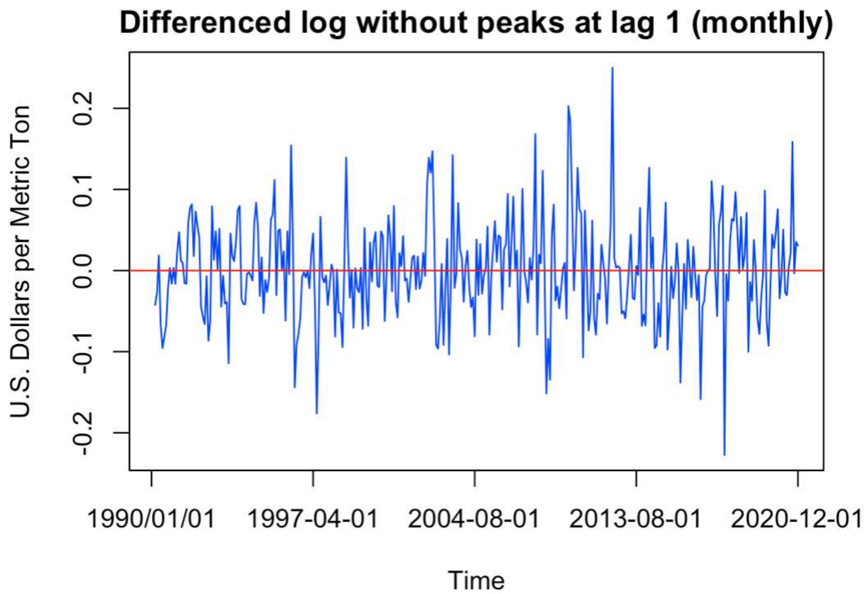

Figure 17: Differenced log-scaled transform of the process excluding 2008 and 2022

```
#Differenced log-scaled transformation without 2008 and 2022 peaks
lv = log(v)
dlv = diff(lv)
plot(dlv, col="blue", xaxt = "n", ylab="U.S. Dollars per Metric Ton", xlab="Time",
    main="Differenced log without peaks at lag 1 (monthly)" )
observational_x_val <- c(0, 88, 176, 264, 352)
custom_values <- c("1990/01/01", "1997-04-01", "2004-08-01", "2013-08-01",
    "2020-12-01")
axis(1, at = observational_x_val, labels = custom_values)
abline(h=0, col="red")
```

As it is clearly exhibited by the second graph (Figure 14), excluding the peaks gives a more stable graph when compared to the differenced log-scaled transform of Figure 5, also displaying a stronger stationarity. This can be explained with the fact that the 2008 and 2022 heavily contributed to the volatility of the process, although the general characteristics of the time series give the impression to have stayed the same and that there are no striking features that have modified.

7 Conclusions

In conclusion, the time series analysis has provided important understandings into the patterns present in our data, specifically the Global Price in U.S. Dollars of Wheat per Metric Ton. After briefly introducing the matter and giving a brief description of the historical reasonings behind increases and decreases of the price, we delved into the preliminary analysis of the time series.
In that section, we decided to take the logarithmic transform of the process in order to attuenate the fluctuations. Afterwards, we analyzed the Autocorrelation and Partial Autocorrelation (ACF and PACF) functions of the process, immediately observing a strong AR component but excluding the possibility of it being the only one.
Taking the differenced log-scaled transform, the data manifested convincing characteristics of stationarity that were further confirmed by the Augmented Dickey-Fuller test, which resulted in a p-value (0.01) that is small enough to be confident in stationarity.
Moreover, the seasonal patterns were analyzed and monthly difference was chosen because it showed less standard deviation than the quarterly and annual differences, along with a lower range and a skewness imminent to zero.
In addiction, we attempted to find a suitable model for our process. After several simulations and the findings of the preliminary analysis, we decided to choose the ARIMA(2,0,1) model with parameters ϕ_1=-0.7, ϕ_2=0.2 and $theta$=1. The chosen ARIMA(2,0,1) model shared with the wheat time series important similarities, such as the differenced log-scaled transform and also remarkable analogies in their Spectral Density Functions, that further confirmed stationality. As the dataset is in continuous transformation, the findings were established considering the prices up to February 2024.

8 Sources

8.1 R Packages

library(tseries)
library(forecast)

8.2 Websites

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https://www.world-grain.com/media/photos/154-world-s-top-wheat-producing-countries

8.3 Books

Time Series: Theory and Methods, Brockwell P.J and Davis R.A
Introduction to Time Series and Forecasting, Second Edition, Brockwell P.J and Davis R.A.